

Classical spins and the dynamical meaning of an operational state

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Abstract

A classical random orientation observed through affine directional responses has the operational state space of a qubit. We use two interacting classical spins to show why this finite statistical description requires a separate dynamical consistency condition. Two preparations with identical retained statistics acquire different terminal probabilities under one smooth reversible Hamiltonian. The interaction admits no exact finite enlargement of the retained expectation coordinates for all probability preparations, even with nonlinear coordinate updates. The explicit example distinguishes statistical dimension, distinguishability capacity and reversible operational closure, and identifies the additional physical input required when applying quantum reconstruction principles to classical mechanics.

1 When a state description predicts its own evolution

An operational state groups preparations that give the same probabilities for an assigned collection of transformations and terminal measurements. Enlarging that collection can split the equivalence classes [1]. Here we start with a quotient defined by terminal tests and local rotations, then ask whether a classical interaction preserves it. A state update on this quotient exists precisely when equal retained states remain equal after the interaction. This tests the sufficiency of the description independently of microscopic reversibility.

For two spins the retained data are their mean orientations and their nine cross moments: $\mathbb{E}u_i$, $\mathbb{E}v_j$ and $\mathbb{E}[u_i v_j]$. These fifteen numbers predict every initial product test in the model. We exhibit two probability preparations with the same fifteen numbers but different later measurement probabilities, then prove that adding finitely many expectation coordinates cannot fix this for all preparations. This is the concrete contribution of the worked example.

Section 2 defines the measurement model and its relation to qubit statistics; Section 3 gives the separating preparations and their explicit motion. Section 4 proves the finite-enlargement obstruction. The counterexample itself needs only classical spin evolution and expectations; the final obstruction also uses finite-dimensional linear algebra.

Restricted classical models and minimal tensor products provide a standard way to separate local statistics from composition [3]. The local sphere-to-ball reduction is an instance of the classical-extension construction reviewed by Bacciagaluppi [2]. Quantum reconstruction results make composition and reversible transformations substantive hypotheses [4, 5, 6]. We illustrate their physical role with a classical spin interaction and an exact observable-closure test. Finite invariant observable spaces are familiar from Koopman theory [8]; the argument here is an explicit application to the retained measurement statistics of this model. Unlike the nonlinear quantum boxes of Ref. [1], our full classical probability measures evolve by linear pushforward; the failure occurs when projecting onto retained moments.

The example uses all Borel probability preparations and stipulated affine terminal measurements. Its result concerns exact finite expectation-coordinate descriptions on a nonzero

time interval. Approximate closure and restricted preparations pose different questions. The fixed spin magnitudes supply action units; the construction leaves the physical selection of a universal action constant as a separate task.

2 A classical orientation with qubit statistics

Let the hidden orientation be $u \in S^2 \subset \mathbb{R}^3$. A preparation is any Borel probability measure μ . An allowed measurement is a finite family

$$e_i(u) = a_i + b_i \cdot u, \quad a_i \geq |b_i|, \quad \sum_i a_i = 1, \quad \sum_i b_i = 0. \quad (1)$$

These nonnegative responses sum to one. Their probabilities depend exactly on $r = \int u d\mu$, since directional binary measurements determine its three components. Every $|r| \leq 1$ is a mixture of opposite orientations. The state space is therefore the unit ball, with three normalized coordinates and four unnormalized linear coordinates.

Using Pauli matrices as a coordinate basis,

$$\rho(r) = \frac{1}{2}(I + r \cdot \sigma), \quad E_i = a_i I + b_i \cdot \sigma, \quad \text{tr}(E_i \rho) = a_i + b_i \cdot r. \quad (2)$$

Thus the allowed local states and effects have the usual qubit representation. Rotations $u \mapsto Ru$, $R \in SO(3)$, descend to $r \mapsto Rr$ and connect all pure ball states continuously and reversibly.

The perfectly distinguishable capacity is two. If outcomes distinguish k states r_i perfectly, then $1 = a_i + b_i \cdot r_i \leq 2a_i$ for each success outcome. Summing gives $k \leq 2$; opposite poles attain two. Capacity counts perfectly distinguishable alternatives in one measurement, whereas statistical dimension counts independent probability coordinates.

2.1 Composition supplies another choice

For n components admit all joint measures on $(S^2)^n$ and product measurements, their mixtures and coarse-grainings. The retained state is

$$z = \int (1, u_1) \otimes \cdots \otimes (1, u_n) d\mu. \quad (3)$$

Product effects span these 4^n unnormalized coordinates. The state space is the convex hull of product ball states, equivalently separable density matrices in the Pauli representation. Local rotations and permutations preserve this description.

Its capacity is 2^n : every outcome is bounded by 2^n times its value at the product of zero ball vectors, including after mixing and coarse-graining. Normalization bounds the number of perfectly distinguished states by 2^n ; product pole tests attain it. Every extreme composite state is product-pure, because the compact product-pure set generates its convex hull. Conversely, pure marginals force each orientation almost surely, making such products extreme. Consequently every marginal of a pure composite is pure, so a mixed local state has no purification in this composite class.

There is also a simple subspace contrast. For two components the equal-sign effect along axis n is

$$e_{\text{same}}(u, v) = \frac{1}{2}[1 + (n \cdot u)(n \cdot v)]. \quad (4)$$

Probability one forces support on (n, n) and $(-n, -n)$. The resulting capacity-two face is a segment rather than an elementary ball. This illustrates the additional content of Hardy's subspace requirement [4] and of purification in the CDP reconstruction [5].

3 A reversible interaction distinguishes equal retained states

Give the spins fixed magnitudes $S_A, S_B > 0$ with action units and choose $J > 0$ with energy units. Standard classical-spin Poisson brackets [7] give

$$\{u_i, u_j\} = S_A^{-1} \epsilon_{ijk} u_k, \quad \{v_i, v_j\} = S_B^{-1} \epsilon_{ijk} v_k, \quad \{u_i, v_j\} = 0, \quad H = J u_z v_z. \quad (5)$$

Use $\dot{f} = \{f, H\}$ and $\Omega_A = J/S_A$, $\Omega_B = J/S_B$. The equations include

$$\dot{u}_x = -\Omega_A u_y v_z, \quad \dot{u}_y = \Omega_A u_x v_z, \quad \dot{u}_z = 0, \quad (6)$$

with the corresponding equations for v obtained by exchanging the spins. Both sphere norms and H are conserved. Smoothness on the compact product of spheres gives a complete flow Φ_t with inverse Φ_{-t} .

Let e_x, e_y, e_z be coordinate unit vectors and prepare

$$\mu_A = \delta_{e_y}(du) \frac{1}{2}(\delta_{e_z} + \delta_{-e_z})(dv), \quad (7)$$

$$\mu_B = \delta_{e_y}(du) \frac{1}{2}(\delta_{e_x} + \delta_{-e_x})(dv). \quad (8)$$

Both have $\mathbb{E}u = e_y$, $\mathbb{E}v = 0$ and $\mathbb{E}uv^T = 0$, so every retained product measurement gives identical initial statistics. Both preparations also have microscopic energy zero.

Since $u_z = 0$, the second spin remains fixed. In μ_A , writing $v = se_z$ with $s = \pm 1$,

$$u(t) = (-s \sin \Omega_A t, \cos \Omega_A t, 0). \quad (9)$$

In μ_B we have $v_z = 0$, so both orientations remain fixed. The allowed product effect $e(u, v) = (1 + u_x)(1 + v_z)/4$ therefore gives

$$p_A(t) = \frac{1}{4}(1 - \sin \Omega_A t), \quad p_B(t) = \frac{1}{4}. \quad (10)$$

For sufficiently small positive time the outputs differ despite identical retained inputs. Thus even a single-valued reduced-state update fails for this interaction and preparation class. The hidden flow remains reversible, and each evolved preparation still defines a valid separable operational state. The missing information appears directly in

$$\frac{d}{dt} \mathbb{E}[u_x v_z] = -\Omega_A \mathbb{E}[u_y v_z^2]. \quad (11)$$

4 Exact finite enlargement also fails

The question is whether storing more averages could make the reduced state sufficient. For exact prediction, every evolved retained observable must have an expectation determined by those stored averages. The first proposition exhibits infinitely many independent evolved observables; the second explains why nonlinear processing of finitely many averages cannot recover them for every preparation.

The pullback $U_t f = f \circ \Phi_t$ acts on bounded Borel observables compared pointwise. All atomic preparations are admitted, so this is stronger than equality modulo a chosen reference measure.

Proposition 1. *No finite-dimensional real space of bounded Borel observables containing 1 and the original terminal effects is invariant under U_t for every time in any nonzero interval, including a forward interval beginning at zero.*

Proof. The complexified space contains $w = u_x + iu_y$. Equation (6) and constancy of v_z give $U_t w = w \exp(i\Omega_A t v_z)$. Suppose finitely many distinct times obey $\sum_{j=1}^m c_j U_{t_j} w = 0$. Restrict to $u = e_x$ and $v = (\sqrt{1-z^2}, 0, z)$, $-1 < z < 1$. Then

$$\sum_{j=1}^m c_j \exp(i\Omega_A t_j z) = 0. \quad (12)$$

Differentiating in z at zero through order $m-1$ gives a Vandermonde system in the distinct numbers $i\Omega_A t_j$. Its determinant is nonzero since $\Omega_A > 0$, so every coefficient vanishes. There are arbitrarily many independent time translates, contradicting finite dimension. Only explicit exponentials were differentiated; added observables need no regularity. $\square$

Proposition 2. *Allowing nonlinear updates of a finite expectation vector does not repair exact evolution for all admitted preparations.*

Proof. Let $f_0 = 1, f_1, \dots, f_d$ be retained real observables. If $\mathbb{E}_\mu g$ is determined by their expectations for every probability measure μ , then g lies in their real linear span. Otherwise evaluation linear algebra gives finitely many points x_j and real weights a_j such that

$$\sum_j a_j f_k(x_j) = 0 \quad (0 \leq k \leq d), \quad \sum_j a_j g(x_j) \neq 0. \quad (13)$$

Indeed, if every relation between the evaluation vectors also annihilated g , evaluation of g would define a linear functional on their span and express g as a combination of the f_k . The relation for $f_0 = 1$ gives equal positive and negative masses. Normalizing them produces two atomic probability measures with identical retained expectations but different g -expectations. Apply this observation to $g = U_t f_k$: any single-valued expectation update forces invariance of the retained span, which the preceding proposition excludes. $\square$

These propositions concern finite statistical dimension with ordinary probabilistic mixing. Infinite observable dimension by itself says nothing about perfectly distinguishable capacity. Approximation, restrictions on preparations and changed dynamics each alter a stated premise. At $J = 0$ the interaction disappears and static finite closure is restored.

5 What the example contributes to reconstruction

De la Torre and colleagues [6] study locally quantum, locally tomographic systems with continuous reversible operational interactions. Their theorem constrains composite structure and, with the specified identical-copy and operational closure assumptions, selects quantum theory. The spin example already has the required local ball representation. Its Hamiltonian interaction instead fails to induce a map on that finite operational state, and finite enlargement cannot supply one for all the admitted preparations. Microscopic reversibility therefore does not discharge the theorem's operational reversibility hypothesis.

Premise	Role in this example
Affine terminal effects	Stipulated restriction yielding the local ball.
All Borel preparations	Makes equality of expectations a pointwise test and permits atomic separation.
Product measurements	Selects the minimal composite and its finite moment coordinates.
Classical spin Hamiltonian	Supplies a complete reversible microscopic evolution.
Exact finite operational closure	Fails under that interaction for the stated preparation class.
Physical action identification	Spin magnitudes are supplied; universality requires additional physical input.

The dimensionless ball construction fixes probability geometry. Mechanical time enters the interaction through $\Omega_{At} = Jt/S_A$, where J and S_A are independently specified model parameters. Relating a reconstructed transformation generator to measured energy and a universal phase per unit action is a further physical identification. The useful conclusion is a precise location for that requirement, alongside the requirement of a finite description stable under the allowed dynamics.

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